! • $t=0$ $\qquad$ $t=?$

^{238}U $\qquad$ 2,00 mg ^{238}U

^{206}Pb $\qquad$ 1,70 mg ^{206}Pb

• $^{238}U \xrightarrow{T=4{,}47\cdot 10^9 j} {}^{206}Pb + \ldots$

? t

$\ddot{O}$: • $^{238}U \rightarrow {}^{206}Pb + \ldots$

1 atoom ^{238}U → 1 atoom ^{206}Pb + ...

1 mol atomen ^{238}U → 1 mol atomen ^{206}Pb + ...

238 g ^{238}U → 206 g ^{206}Pb

$\frac{238}{206}$ g ^{238}U → 1 g ^{206}Pb

$1{,}7 \cdot \frac{238}{206}$ mg ^{238}U → 1,70 mg ^{206}Pb

1,964 mg ^{238}U → 1,70 mg ^{206}Pb

• $N(t) = N_0 \cdot 2^{-t/T}$

$n(t) \cdot \cancel{N_A} = n_0 \cdot \cancel{N_A} \cdot 2^{-t/T}$

$\frac{m(t)}{\cancel{M}} = \frac{m_0}{\cancel{M}} \cdot 2^{-t/T}$

$\underbrace{2}_{=Y_1} = \underbrace{3{,}964 \cdot 2^{-t/4{,}47\cdot 10^9 j}}_{=Y_2}$

TBLSTART = 0

ΔTBL = 1E8

$\Rightarrow t = 4{,}4 \cdot 10^9$ à $4{,}5 \cdot 10^9$ j.

$\boxed{D = OK}$